

Ricky Faure

Principal Software Engineer

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SUMMARY

I build scalable enterprise-grade platforms with cutting edge open source tooling. Experienced in Kubernetes hosted applications, Operator Development, CI/CD, Terraform, and Distributed Systems.

SKILLS

Core: Kubernetes Go Terraform GCP

Additional: Kubebuilder / Operator SDK Helm Kafka Elasticsearch GitHub Actions Python CI/CD GitOps
Distributed Systems GKE Docker Prometheus Java Spring Boot Baremetal Kubernetes IaC

EXPERIENCE

Optum

2019 — PRESENT

Principal Software Engineer, TLCP — S.E. → Sr. S.E. → Lead S.E. → Principal S.E.

Empowering hundreds of teams across the enterprise to build data-driven services and moving tens of petabytes of data with minimal architectural overhead — using Kubernetes Operators and CI/CD to automate deployment, management, and hosting of enterprise-grade distributed systems across on-premise and multi-cloud environments.

- Led a team of 7 engineers building Kafka-as-a-Service and Elasticsearch-as-a-Service platforms, achieving 99.5% reliability via Kubernetes Operators and robust infrastructure-as-code practices.
- Built custom Kubernetes Operators for Kafka, Elasticsearch, Prometheus, Service Monitors, GCP VolumeSnapshots, Certificate Management, and bare-metal Kubernetes upgrade orchestration.
- Achieved a 52% increase in resource deployments by migrating from GitOps to a GUI-based management system within an enterprise marketplace, backed by a custom Terraform Provider and Kubernetes resource manager.
- Delivered \$2.5M in annual cost savings through cloud resource optimization, instance type migrations, and elimination of excessive log retention in GCP.
- Expanded platform to handle 20+ petabytes of data movement across on-premise and GCP (GKE, GAR, IAM, GCS, VPC) by extending Kubernetes Operators for cloud environments.
- Designed a vulnerability scanning workstream using reusable GitHub Actions, a custom Python Rally API library, and automated report generation — adopted org-wide.
- Mentored junior engineers in rotational programs; taught quarterly courses on Kubernetes Operators and Elasticsearch/Kibana.

Technologies: Go, Kubebuilder, Kubernetes Operators, Helm, Terraform, GCP, GitHub Actions, Python, Kafka, Elasticsearch, Jenkins

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INTERNSHIP

TDP Software Development Intern

Contributed business-critical features to a monolithic application, complete with feature implementations and automated unit and integration testing.

Technologies: Java, Spring Boot, Groovy Spock

PROJECTS

Run Samurai, Run!

P5.js · HTML/CSS · PHP · AJAX · Game Loop Dev

Top-down 2D infinite side-scrolling runner for mobile. Players fight through enemy mobs for upgrades and high scores, with single-player and head-to-head multiplayer modes.

SafetyNet

Swift · iOS · P2P Networking · E2E Encryption

iOS peer-to-peer messaging app requiring no Wi-Fi or cellular data. Uses device-to-device networking with end-to-end encryption for resilient off-grid communication.

Rubik's Cube Genetic Algorithm Solver

Python · Genetic Algorithms · Matrix Math

Evolutionary algorithm that solves a fully simulated Rubik's Cube model — exploring genetic computation applied to combinatorial optimization.

Ray Tracing Engine

Vanilla JS · Graphics Pipeline · 3D Rendering · Blinn-Phong

Browser-based ray tracer in vanilla JavaScript. Renders spheres and triangles with Blinn-Phong lighting, reflections, and a free-moving camera — no GPU required.

Sheet Vision— Senior Design

ElectronJS · ReactJS · AWS · Python · Computer Vision

Application that reads sheet music, plays it back, and listens to the user in real time — providing feedback to help learners draw parallels between notation and sound.